

Lecture 7 — Static and Dynamic Revenue Management at Scale

Lecturer: Sid Banerjee

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7.1 From Littlewood to interleaved arrivals

Lecture 6 studied a separated-arrival model: low-fare requests arrived first and high-fare requests arrived later. With fares $0 < r_L < r_H$, protecting the y th unit for later high-fare demand D_H was worthwhile when

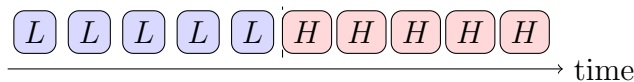
$$r_H \mathbb{P}(D_H \geq y) \geq r_L.$$

Thus a Littlewood protection level can be written, with ties settled in favor of protection, as

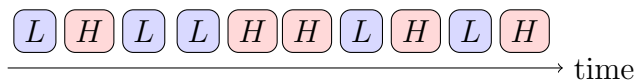
$$y^* = \max \left\{ 0 \leq y \leq C : \mathbb{P}(D_H \geq y) \geq \frac{r_L}{r_H} \right\}. \quad (7.1)$$

The economic comparison was current low-fare revenue versus the expected displacement value of capacity.

Littlewood: separated classes



Lecture 7: interleaved classes



What changes when the two fare classes are mixed over time? Our central question is

How much do dynamic decisions gain over a simple static policy?

Dynamic programming identifies the optimal policy. Our main theorem asks a different question: how close can a fixed protection level come when the market is large?

7.2 The two-fare model: static versus dynamic policies

There are T periods and initial capacity C . In period t , an arrival type $J_t \in \{H, L, 0\}$ is drawn independently with

$$\mathbb{P}(J_t = H) = q_H, \quad \mathbb{P}(J_t = L) = q_L, \quad \mathbb{P}(J_t = 0) = q_0 = 1 - q_H - q_L.$$

A high request earns r_H , a low request earns r_L , and $0 < r_L < r_H$. Each accepted request consumes one unit. The seller observes the fare class before deciding whether to accept it.

This is an *observed-class admission model*: the fare class is known when the accept/reject decision is made. It continues Littlewood directly and differs from choosing a posted price before observing whether a customer buys.

Let the total class demands be the random variables

$$D_H = \sum_{t=1}^T \mathbf{1}\{J_t = H\}, \quad D_L = \sum_{t=1}^T \mathbf{1}\{J_t = L\}.$$

7.2.1 What “static” means here

Fix an integer protection level $y \in \{0, 1, \dots, C\}$. The static protection policy π_y :

- accepts every high-fare request whenever capacity remains;
- accepts a low-fare request whenever capacity remains and fewer than $C - y$ low-fare requests have previously been accepted;
- rejects every request after capacity is exhausted.

Equivalently, $C - y$ is a fixed low-fare booking limit. The control is static because y is chosen before arrivals and is never recomputed.

A dynamic policy instead adapts to remaining time, remaining capacity, and observed history.

7.3 Exact dynamic programming

Let $v_t(b)$ be the maximum expected future revenue with t periods and b units of capacity remaining. Our time index counts down. The boundary conditions are

$$v_0(b) = 0, \quad v_t(0) = 0.$$

For $t, b \geq 1$, the Bellman equation is

$$\begin{aligned} v_t(b) = & q_0 v_{t-1}(b) \\ & + q_H \max\{v_{t-1}(b), r_H + v_{t-1}(b-1)\} \\ & + q_L \max\{v_{t-1}(b), r_L + v_{t-1}(b-1)\}. \end{aligned} \quad (7.2)$$

Inside each maximum, rejection leaves value $v_{t-1}(b)$, while accepting class j earns $r_j + v_{t-1}(b-1)$. Define the marginal continuation value

$$\delta_{t-1}(b) = v_{t-1}(b) - v_{t-1}(b-1).$$

Then a class- j request is accepted precisely when

$$r_j \geq \delta_{t-1}(b). \quad (7.3)$$

This is the same opportunity-cost principle as Littlewood's rule. The difference is that the exact opportunity cost now depends on both remaining time and remaining capacity.

7.4 The scaling question

Solving (7.2) identifies the optimal dynamic policy, but it does not show how much revenue the simpler static policy loses. Suppose the horizon grows, capacity satisfies $C = cT$ for fixed $c \in (0, 1)$, and fares and arrival probabilities remain fixed. We focus on the balanced regime

$$q_H < c < q_H + q_L, \quad (7.4)$$

so expected high demand alone does not fill capacity, but expected total demand does. We will show that the Littlewood policy loses at most $O(\sqrt{T})$ against hindsight, and that this order is tight for that policy on a simple instance.

7.5 The single-resource hindsight benchmark

Let OPT be the random revenue of an offline clairvoyant that knows all arrivals in advance. On every sample path it serves high-fare demand first and then uses remaining capacity for low-fare demand:

$$\text{OPT} = r_H \min\{C, D_H\} + r_L \min\{(C - D_H)^+, D_L\}. \quad (7.5)$$

Let R_y be the random revenue of static policy π_y . Since hindsight has more information than the dynamic policy, and the optimal dynamic policy dominates any static policy,

$$\boxed{\mathbb{E}[\text{OPT}] \geq v_T(C) \geq \mathbb{E}[R_y]}. \quad (7.6)$$

In particular,

$$v_T(C) - \mathbb{E}[R_y] \leq \mathbb{E}[\text{OPT} - R_y]. \quad (7.7)$$

An upper bound against hindsight therefore also bounds the loss relative to the optimal dynamic policy.

7.6 Demand fluctuations and the square-root law

The high-fare count is binomial, with

$$d_H := \mathbb{E}[D_H] = Tq_H, \text{ and } \text{Var}(D_H) = Tq_H(1 - q_H).$$

Therefore Cauchy–Schwarz gives

$$\mathbb{E}|D_H - d_H| \leq \sqrt{Tq_H(1 - q_H)}. \quad (7.8)$$

This is the same concentration phenomenon seen in Lectures 4–5, viewed at two scales:

average of T observations : uncertainty of order $1/\sqrt{T}$, sum of T observations : uncertainty of order $\sqrt{T}$.
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In UCB, the unknown reward distribution must be learned. Here the distribution is known, but the future realization is not. Random demand still fluctuates by order $\sqrt{T}$.

7.7 Performance of a static protection policy

We first prove a bound for any fixed protection level y .

Theorem 7.1 (Sample-path comparison). *On every realized sample path,*

$$0 \leq \text{OPT} - R_y \leq (r_H - r_L)(D_H - y)^+ + r_L(y - D_H)^+. \quad (7.9)$$

Consequently, with $\kappa = \max\{r_H - r_L, r_L\}$,

$$\mathbb{E}[\text{OPT} - R_y] \leq \kappa \left(\sqrt{Tq_H(1 - q_H)} + |d_H - y| \right). \quad (7.10)$$

Proof: If $D_H < y$, the policy has enough protected capacity to accept every high request. It accepts at most $C - y$ low requests, while hindsight can accept at most $C - D_H$. Thus hindsight can add at most $y - D_H$ low requests, for gain no larger than $r_L(y - D_H)$.

If $D_H \geq y$, the booking limit leaves room for at least y high requests. Any additional high request rejected because earlier low requests filled capacity can replace at most one accepted low request in hindsight. At most $D_H - y$ replacements are needed, and each improves revenue by $r_H - r_L$. This proves (7.9).

The right side is at most $\kappa|D_H - y|$. The triangle inequality and (7.8) give

$$\mathbb{E}|D_H - y| \leq \mathbb{E}|D_H - d_H| + |d_H - y| \leq \sqrt{Tq_H(1 - q_H)} + |d_H - y|.$$

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Now use the Littlewood level from (7.1), computed for $D_H \sim \text{Binomial}(T, q_H)$. Let

$$\alpha = \frac{r_L}{r_H} \in (0, 1), \quad m = \min\{\alpha, 1 - \alpha\}.$$

Because y^* is a fixed tail quantile, Chebyshev's inequality and maximality in (7.1) give the coarse but sufficient estimate

$$|y^* - d_H| \leq 1 + \frac{\sqrt{Tq_H(1 - q_H)}}{\sqrt{m}}. \quad (7.11)$$

Under (7.4), $y^* < C$ for all sufficiently large T . Combining the bounds yields the main result.

Corollary 7.2 (Littlewood policy at scale). *For fixed r_H, r_L, q_H, q_L , and c satisfying (7.4),*

$$\mathbb{E}[\text{OPT} - R_{y^*}] \leq \kappa \left[1 + \left(1 + \frac{1}{\sqrt{m}} \right) \sqrt{Tq_H(1 - q_H)} \right] = O(\sqrt{T}). \quad (7.12)$$

Hence also $v_T(C) - \mathbb{E}[R_{y^*}] = O(\sqrt{T})$.

The estimate is deliberately coarse: Chebyshev bounds the relevant tail distance by a constant multiple of the standard deviation, and the extra 1 accounts for integer indexing.

7.8 A square-root lower-bound example

The upper bound is tight for this particular static Littlewood policy in a simple instance. Let T be even and set

$$C = T/2, \quad q_H = q_L = 1/2, \quad q_0 = 0, \quad r_H = 2, \quad r_L = 1.$$

Then $D_H \sim \text{Binomial}(T, 1/2)$ and $D_L = T - D_H$. Since $r_L/r_H = 1/2$, symmetry gives $y^* = T/2 = C$. The low-fare booking limit is zero, so the static policy rejects every low request and earns

$$R_{y^*} = 2 \min\{D_H, C\}.$$

Hindsight fills any high-demand shortfall with low requests, so

$$\text{OPT} - R_{y^*} = (C - D_H)^+. \quad (7.13)$$

The standard deviation of D_H is $\sqrt{T}/2$. By the central limit theorem, negative deviations proportional to $\sqrt{T}$ occur with constant probability. Hence

$$\boxed{\mathbb{E}[\text{OPT} - R_{y^*}] = \Theta(\sqrt{T})}. \quad (7.14)$$

This is a matching lower bound for the hindsight regret of this particular static Littlewood policy. It is *not* a proof that

$$v_T(C) - \mathbb{E}[R_{y^*}] = \Omega(\sqrt{T}).$$

because hindsight may itself exceed the optimal online dynamic value by order $\sqrt{T}$. A dynamic-versus-static lower bound requires a separate argument.

7.9 First-order optimality and what comes next

In the scaling regime, revenues are typically $\Theta(T)$ while the static policy's additive hindsight loss is only $O(\sqrt{T})$. Its relative loss is therefore $O(1/\sqrt{T})$. The policy is *first-order optimal*: scale makes a simple static control accurate as a fraction of market value. This does not say that static and dynamic policies are identical.

A dynamic policy can relax protection after unexpectedly weak high-fare demand or tighten it after unexpectedly strong demand. With multiple shared resources, even one adaptive scalar protection level is no longer enough: a request may consume several resources simultaneously. The next lecture develops fluid LPs, dual resource prices, and bid-price control for this network setting.

7.10 Takeaways

1. Littlewood separates fare classes in time; the interleaved model requires decisions throughout the horizon.
2. Exact dynamic programming prices capacity using both remaining time and remaining inventory.
3. Hindsight, the dynamic optimum, and a static policy are distinct benchmarks.
4. The Littlewood protection level has $O(\sqrt{T})$ hindsight regret in the balanced scaling regime, and a specific instance makes this bound tight for that policy.
5. Because revenue is order T , an order- $\sqrt{T}$ loss is first-order negligible. The same square-root fluctuation underlies the $1/\sqrt{T}$ estimation scale from UCB.

Recommended reading

Talluri and van Ryzin, Sections 2.5.1–2.5.2, for dynamic programming. The static-policy comparison and square-root bounds are developed directly in these notes.